

MODEL QUESTION PAPER

REVISION-2010  
SUB CODE: 3080

Reg. No.....  
Signature.....

THIRD SEMESTER DIPLOMA EXAMINATION IN CHEMICAL ENGINEERING

**POLYMER TECHNOLOGY**

Time : 3 Hour

Max. marks:100

**PART A**

(Answer all question . Each question carry 2 marks )

1. Differentiate thermo plastics and thermo setting plastics.
2. State different types of polymerization reactions.
3. What are the additives used in plastics?.
4. What are the moulding techniques used in plastic manufacture?
5. What is number average molecular weight?

**PART B**

(Answer any five questions Each question carry 6 marks )

- |   |                                                                             |   |
|---|-----------------------------------------------------------------------------|---|
| 1 | How to classify polymers according to their properties                      | 6 |
| 2 | Explain characters features and application of co-polymerization            | 6 |
| 3 | What are the material used for reinforcement                                | 6 |
| 4 | What is meant by calendaring                                                | 6 |
| 5 | Explain the preparation method of urea formaldehyde                         | 6 |
| 6 | What are the characteristics of poly condensation                           | 6 |
| 7 | Explain the manufacture of malamine formaldehyde resin and mention its uses | 6 |

## PART C

(Answer one full question from each module. Each question carries 15 marks)

|     |                                                                                       |   |
|-----|---------------------------------------------------------------------------------------|---|
| I.  | MODULE I                                                                              |   |
| a   | Explain the manufacture of caprolactum                                                | 4 |
| b   | Explain the manufacture of propylene and its uses                                     | 4 |
| c   | Explain the copper wire test for analyzing polymer material                           | 3 |
| d   | Explain the mechanical properties of polymers                                         | 4 |
|     | OR                                                                                    |   |
| II  |                                                                                       |   |
| a   | Explain the determination of molecular weight of polymer material by viscosity method | 6 |
| b   | Explain the manufacture of vinyl chloride and its uses                                | 6 |
| c   | State the polymer morphology                                                          | 3 |
|     | MODULE II                                                                             |   |
| III |                                                                                       |   |
| a   | Write a short note on addition polymerization                                         | 4 |
| b   | Explain the different types of copolymers                                             | 4 |
| c   | What are the characteristics of polycondensation                                      | 4 |
| d   | What is meant by step polymerization                                                  | 3 |
|     | OR                                                                                    |   |
| IV  |                                                                                       |   |
| a   | Explain the addition polymerization with examples                                     | 6 |
| b   | Explain about chain polymerization with examples                                      | 6 |
| c   | What is meant by graft copolymer?                                                     | 3 |
|     | MODULE III                                                                            |   |
| V   |                                                                                       |   |
| a   | What are the commodity plastics                                                       | 3 |
| b   | Explain the manufacture of polypropylene                                              | 6 |
| c   | Write short notes on polyurethane                                                     | 4 |
| d   | What are the functions of antioxidants                                                | 2 |
|     | OR                                                                                    |   |
| VI  |                                                                                       |   |
| a   | Explain the manufacture of phenol formaldehyde and its uses                           | 6 |
| b   | Explain the manufacture of urea formaldehyde and its uses                             | 6 |
| c   | Write a short note on plasticizers                                                    | 3 |
| VII | MODULE IV                                                                             |   |
| a   | Explain the manufacture of viscous rayon                                              | 6 |
| b   | Explain die casting                                                                   | 4 |
| c   | Explain short note on film casting                                                    | 5 |
|     | OR                                                                                    |   |
| a   | Describe the manufacture of polyester                                                 | 6 |
| b   | Short note on the following                                                           | 3 |
|     | 1) Melt spinning                                                                      | 3 |
|     | 2) Solution spinning                                                                  | 3 |
|     | 3) Dry spinning                                                                       |   |